

DramValue - Technical Overview

Secondary Market Whisky Price Intelligence Platform

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Overview

DramValue is a free whisky price tracking platform that aggregates auction data from multiple sources to help collectors and enthusiasts understand secondary market values. The platform provides real-time price data, trend analysis, and collection management tools.

Technology Stack

Backend Framework

- **FastAPI** (Python 3.11+) - Modern async web framework
- **Uvicorn** - ASGI server with hot reload for development
- **Jinja2** - Server-side HTML templating

Database

- **PostgreSQL 16** - Primary relational database
- **SQLAlchemy 2.0** - Async ORM with type hints
- **Alembic** - Database migrations
- **asyncpg** - High-performance async PostgreSQL driver

Authentication & Security

- **JWT Tokens** - Stateless authentication
- **HTTP-only Cookies** - Secure token storage for SSR
- **bcrypt** - Password hashing
- **python-jose** - JWT encoding/decoding

Frontend

- **Tailwind CSS** - Utility-first styling
- **Alpine.js** - Lightweight reactive JavaScript
- **Chart.js** - Price history visualizations

Data Collection

- **Scrapy** - Web scraping framework
- **Playwright** - Headless browser automation for JavaScript-heavy sites
- **Tor Proxy** - IP rotation for anti-bot detection

Background Processing

- **Celery** - Distributed task queue
- **Redis** - Message broker and caching
- **Celery Beat** - Scheduled task runner

Data Science & ML

- **Pandas/NumPy** - Data manipulation
- **Prophet** - Time series forecasting
- **scikit-learn** - Anomaly detection
- **RapidFuzz** - Fuzzy string matching for bottle normalization

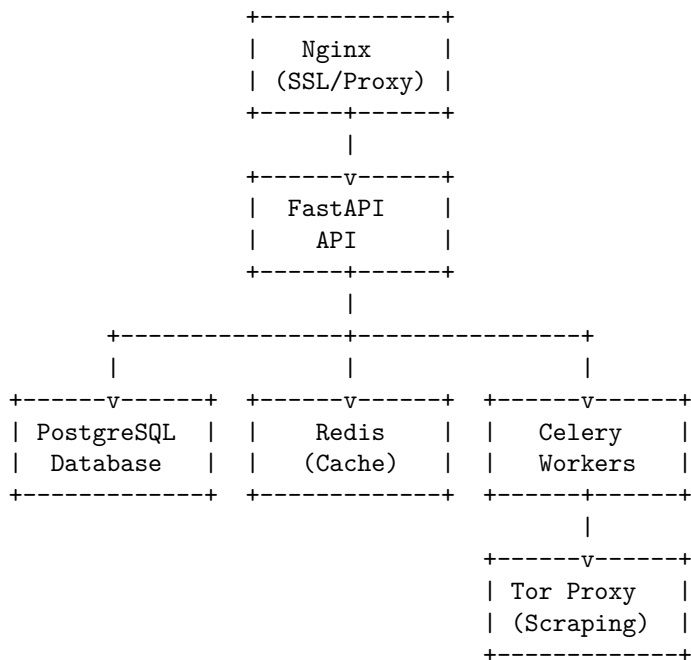
AI Integration

- **Perplexity AI** - Enhanced search and recommendations

Infrastructure

- **Docker Compose** - Container orchestration
- **Nginx** - Reverse proxy and SSL termination

Architecture



Key Features

Price Tracking

- Real auction results from multiple sources
- Historical price charts
- Average, min, max price statistics

User Features

- Personal bottle collections
- Price alerts for target prices
- Portfolio value tracking

Data Sources

- Scotch Whisky Auctions
- Whisky Auctioneer
- Dekanta (retail)
- The Whisky Barrel (retail)

Database Schema

Core models include:

- **Bottles** - Master bottle catalog with normalization
- **Prices** - Individual price records with source tracking
- **Users** - Authentication and preferences
- **Collections** - User bottle portfolios
- **CollectionItems** - Bottles in collections with purchase info
- **PriceAlerts** - User-defined price notifications

Security Considerations

- CORS restricted to dramvalue.com domain
- HTTP-only cookies prevent XSS token theft
- Parameterized queries prevent SQL injection
- Rate limiting on authentication endpoints
- Tor proxy rotation avoids IP bans during scraping

Development Workflow

1. Docker Compose manages all services
2. Hot reload enabled for rapid development
3. Alembic handles database migrations
4. pytest for automated testing
5. ruff for code linting and formatting

Built with modern Python and a focus on performance, security, and user experience.